Skills Summary

Indexes, Ratios and Convergence

Use this reference while completing your worksheet or when studying.

Skill 1 — Counting terms and reading the index

The index is data, not decoration. A rule written for $n \geq 1$ puts the first term at $n = 1$, so the kth term is the value at $n = k$. Substituting $k - 1$ gives the term *before* the one you wanted.

In a sum, read the limits. $\sum_{k=1}^{4}$ has four terms and no $k = 0$ term; $\sum_{k=0}^{4}$ has five. Count the terms before you add: from $k = p$ to $k = q$ there are $q - p + 1$ of them.

Example: For $a_n = 3n + 5$ with $n \geq 1$, find the fourth term.
Step 1. The rule starts at $n = 1$, so "fourth term" means the value at $n = 4$ — no shifting is needed or allowed.
Step 2. $a_4 = 3(4) + 5 = 12 + 5$
$\Rightarrow$ **17** (the value at $n = 3$ would be 14, the *third* term)

Try it: For $a_n = 7n - 2$ with $n \geq 1$, find the sixth term.

check: **40**

Skill 2 — Common ratio and geometric sums

Find the ratio by dividing, never subtracting: $r = a_{n+1}/a_n$, and it must come out the same for every consecutive pair.

Finite sum: $S_n = a_1(1 - r^n)/(1 - r)$. **Infinite sum:** $S = a_1/(1 - r)$, *but only when* $-1 < r < 1$. Outside that range the terms do not shrink, the series has no sum, and the formula returns a meaningless number.

Example: For $8 + 4 + 2 + 1 + \cdots$, find the ratio, the sum of the first four terms, and the infinite sum.
Step 1. Dividing consecutive terms gives the ratio, and its size decides whether the infinite sum formula may be used at all.
Step 2. $r = 4/8$, and that ratio lies between -1 and 1, so both formulas apply.
Step 3. $S_4 = 8\big(1 - (1/2)^4\big)/(1 - 1/2) = 8(15/16)(2)$, and $S = 8/(1 - 1/2) = 8(2)$.
$\Rightarrow$ $r = \mathbf{\frac{1}{2}}$, $S_4 = \mathbf{15}$, $S = \mathbf{16}$

Try it: For $9 + 3 + 1 + \cdots$, find the common ratio and the infinite sum.

check: $r = \mathbf{\frac{1}{3}}$, $S = \mathbf{\frac{27}{2}}$

(!) Watch out: for $3 + 6 + 12 + \cdots$ the ratio is 2, so there is no infinite sum. Writing $3/(1 - 2) = -3$ is using a formula outside the range where it means anything — and a sum of positive terms can never be negative.

Skill 3 — Binomial expansions term by term

The term rule: in $(A+B)^n$ written in descending powers of A, the term at index k is $\binom{n}{k}A^{n-k}B^k$, and k starts at 0. So the *third* term uses $k=2$, not $k=3$.

Square the whole first piece. In $(2x+3)^5$ the A is $2x$, so A^2 is $4x^2$ — the coefficient is squared along with the letter.

Example: Expand $(x+3)^3$.
Step 1. Row 3 of Pascal's triangle supplies the coefficients, and the powers of 3 rise as the powers of x fall, so the whole expansion can be written straight down.
Step 2. Coefficients $1, 3, 3, 1$ with $3^0, 3^1, 3^2, 3^3$: $x^3 + 3x^2(3) + 3x(9) + 27$.
$\Rightarrow \quad \boldsymbol{x^3 + 9x^2 + 27x + 27}$

Try it: Expand $(x+1)^4$.

check: $\boldsymbol{x^4 + 4x^3 + 6x^2 + 4x + 1}$